INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

(School of University of Engineering and Management, Kolkata)

End Semester Examination, April – May, 2025

Degree: **B.Tech**, Semester: **Second**

Stream: **Biotech/CSE(AI)/CSE(AI & ML)/ CSE(AIML)/CSE(Cyber Security)/CSE(Data Science)/CSE(IOT) /CSE(IOT, CS, BT)/ CSE(IOTCSBT)/CSE(Networks) /Robotics and AI/CSE/CSIT/CST/IT/CSBS/ECE/ME/EE/EEE**

Paper Code: **BSCM203**

Paper Name: **Mathematics - II**

Full Marks: **100** Duration: **3 Hours**

Part - A

Attempt any 10 questions out of 15 questions

Each question carries 2 Marks (2 × 10)

1. Evaluate the following integral: $\int_1^2 \int_1^3 xy^2 dxdy$ **2**

2. Evaluate the following integral: $\int_0^2 \int_0^x (x+y) dxdy$ **2**

3. Evaluate the following integral: $\int_0^\pi \int_0^{asin\theta} rdrd\theta$ **2**

4. If $\mathbf{F} = (5xy - 6x^2)\mathbf{I} + (2y - 4x)\mathbf{J}$, evaluate $\int_C \mathbf{F} . d\mathbf{R}$ along the curve C in the xy-plane, $y=x^3$ from the point (1,1) to (2,8). **2**

5. **Solve:** $(5x^4+3x^2y^2-2xy^3)dx+(2x^3y-3x^2y^2-5y^4)dy = 0$ **2**

6. Determining the linearity, order, and degree of the given differential equation $(\frac{dx}{dy})^2 + 5y^{1/3} = x.$ **2**

7. Determining the integrating factor of the given differential equation $xy' + y = x^3y^6$. **2**

8. Determining the P.I. of the differential equation $(D^2+D+1)y= \sin 2x$. **2**

9. Show that the set {(0,x,y):x,y are reals} is a subspace of R^3 . **2**

10. Find whether the vectors (2,4,0),(0,1,0) and (2,6,2) are linearly independent in the real vector space R^3. **2**

11. Let V= Set of all continuous functions defined on [a, b] interval. Define the transformation $T: V \to R \; by \; T(f) = \int_a^b f(t)dt$. Prove that T is a linear transformation. **2**

12. Find whether the following transformations are linear or not. **2**
$T: R^3 \to R^2$ defined by $T(x_1, x_2, x_3) = (|x_1|, 0)$

13. If all the letters of word 'ENGINEER' be written at random, what is the probability that all the latters E are found together. **2**

14. Find the probability that a non-leap year should have 53 Saturdays? **2**

15. Find the mean, median and mode for the following: **2**

Mid Value:	15	20	25	30	35	40	45	50	55
Frequency:	2	22	19	14	3	4	6	1	1

Part - B
Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. Show that $\iint_R r^2 \sin\theta \, dr d\theta = 2a^2/3$, where R is the semi-circle $r = 2a\cos\theta$ above the initial line. **5**

17. Find the area of the portion of the cylinder $x^2+y^2=4y$ lying inside the sphere $x^2+y^2+z^2=16$. **5**

18. Evaluate $\int_C (xy + z^2) ds \; where \; C \; is \; the \; arc \; of \; the \; helix \; x = \cos t, y = \sin t, z = t$ which joins the points (1,0,0) and (-1,0,π). **5**

19. Solving the given first-order linear differential equation for *y*. **5**

$$(1 - x^2)\frac{dy}{dx} + 2xy = x\sqrt{(1 - x^2)}$$

20. Solving the second-order non-homogeneous linear differential equation **5**
$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 6e^{3x} + 7e^{-2x} - log2$

21. A linear transformation $T: R^3 \rightarrow R^2$ is such that T(i)= (0,0). **5**
T(j)= (1,1), T(k)= (1,-1) where
i= (1,0,0), j= (0,1,0), k= (0,0,1).
Use the basis $\{i,j,k\}$ in R^3 and the basis $\{w_1,w_2\}$ in R^2 where w_1=(1,1), w_2=(1,2). Determine the matrix of T relative to these basis. Use this matrix to find T(3,1,1).

22. On the space $M_2(R)$ (set of all 2nd order squatre matrix of real number). **5**
<A,B> is defined as
<A,B>= Tr(AB) (trace of the matrix)
Where $A, B \in M_2(R)$. Find whether this is an inner product.

23. A purse contains 2 silver and 4 copper coins and a second purse contains 4 silver and 4 copper coins. If a coin is selected at random from one of the two purses, what is the probability that it is a silver coin? **5**

24. If the coefficient of correlation between two variables x and y is 0.5 and the acute angle between their lines of regression is $\tan^{-1}(3/8)$, show that **5**
$\sigma_x = \frac{1}{2}\sigma_y$.

Part - C
Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. Show that $\iiint \frac{dxdydz}{\sqrt{(a^2-x^2-y^2-z^2)}} = \frac{\pi^2 a^3}{8}$, the integral being extended for all the values of the variables for which the expression is real. **10**

26. a) Verify Green's theorem for $\int_C [(3x-8y^2)dx + (4y-6xy)dy]$ where C is the boundary of the region bounded by x=0,y=0 and x+y=1. **10**
b) Evaluate by Stoke's theorem $\oint_C (yz\,dz + zx\,dy + zx\,dz)$ where C is the curve $x^2 + y^2 = 1, z = y^2$.

27. Find the complete solution of y''-2y'+2y=x+e^{x}cos x. **10**

28. Solve $y = 2px + \gamma^2 p^3$, where $p = \frac{dy}{dx}$ **10**

29. Use Gram-Schmidt process to convert the basis {(1,2,-2),(2,0,1),(1,1,0)} of R^3 into an orthogonal basis and next to an orthonormal basis. **10**

30. A linear transformation $T: R^2 \rightarrow R^3$is such that **10**

T (1,0)=(1,0,1),T(0,1)=(-1,0,1). Determine

a) T ($2i$-$3j$), Nullity; Rank of T.

b) The matrix of T.

31. A student takes his examination in four subjects *P,Q,R,S*. He estimates his chances of passing in *P* as 4/5 , in *Q* as 3/4 , in *R* as 5/6 and in *S* as 2/3. To qualify, he must pass in *P* and at least two other subjects. What is the probability that he qualifies? **10**

32. Calculate the rank correlation coefficient form the following data showing ranks of 10 students in two subjects: **10**

Maths:	3	8	9	2	7	10	4	6	1	5
Physics:	5	9	10	1	8	7	3	4	2	6

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